

Path-Dependent Volatility with Echo State Networks

Repository summary and numerical-validation status

Project summary generated from the repository sources and protocol reports

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Abstract

This repository is the numerical-validation component of a research programme on whether reservoir computing can provide a tractable model of path-dependent stochastic volatility. Its central object is ESN A2-981003, a two-layer echo-state-network volatility model. Five standalone Python protocols generate paths from known stochastic-volatility data-generating processes (DGPs), calibrate the ESN and benchmark models with a common multi-statistic objective, and save figures and LaTeX reports. The work provides encouraging, but not conclusive, evidence that the ESN can reproduce several market stylised facts. In particular, the numerical architecture selection and some fitted read-out effects remain noisy and require higher-budget multi-seed confirmation.

1 What is in the repository

There is no application package or automated test suite. The five independent scripts each contain simulation, statistics, calibration, plotting, and report-writing code; their tracked PNG, JSON, `.tex`, and PDF outputs document historical runs.

Folder	DGP / reference	Question tested
GridSearch_ HyperParameters_ Tuning	fBM log-normal SV (rough and persistent), MRW, rough Bergomi	Shared ESN hyperparameters and reservoir dimensions
Roughness	fBM log-normal SV	Hurst roughness, covariance scaling, monofractality
Taylor_Effect	Multifractal Random Walk (MRW)	Whether $\text{ACF}(r) > \text{ACF}(r^2)$
Leverage_Effect Zumbach_Effect	Strong-leverage Heston 4-factor Guyon–Lekeufack PDV	Negative $\text{Corr}(r_t, V_{t+L})$ Time-reversal asymmetry of volatility/magnitude

All comparisons use Gaussian innovations. Except in the Zumbach protocol, which uses PDV–GL as its DGP and therefore compares ESN with QRH only, the candidate models are the ESN, quadratic rough Heston (QRH), and the two-factor path-dependent-volatility model of Guyon–Lekeufack (PDV–GL).

2 Model under study: a finite-dimensional path-dependent reservoir

The ESN combines a fast/slow linear reservoir $r \in \mathbb{R}^{N_r}$ with a nonlinear feedback bank $z \in \mathbb{R}^{N_z}$. The r modes have geometrically spaced mean-reversion rates and share one innovation $\varepsilon_t \sim \mathcal{N}(0, 1)$:

$$\lambda_i = \lambda_{\text{lo}} \left(\frac{\lambda_{\text{hi}}}{\lambda_{\text{lo}}} \right)^{\frac{i-1}{N_r-1}}, \quad r_{i,t} = e^{-\lambda_i \Delta t} r_{i,t-1} + \sqrt{1 - e^{-2\lambda_i \Delta t}} \varepsilon_t. \quad (1)$$

Their common shock makes $q^\top r_t$ a sum-of-exponentials, Markovian approximation to a Volterra/fractional memory kernel. The normalized read-out direction is based on the calibrated internal memory exponent H_{ESN} :

$$\tilde{q}_i = \lambda_i^{1/2 - H_{\text{ESN}}}, \quad q = \frac{\tilde{q}/\|\tilde{q}\|_2}{\sqrt{(\tilde{q}/\|\tilde{q}\|_2)^\top C(\tilde{q}/\|\tilde{q}\|_2)}}, \quad q^\top C q = 1. \quad (2)$$

Here H_{ESN} is *not* the DGP Hurst parameter; it is an ESN calibration variable selected to match the realised roughness statistic.

The slower z bank is a nonlinear filter of fast $\times$ slow r taps, with a **tanh** activation, even (variance-like), linear (leverage-like), self, and neighbouring-mode terms. This construction is intended to give the model a controlled source of nonlinear multiscale and Zumbach-like effects. The volatility read-out is

$$\eta_t = b_0 + \rho_r a_r (q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z} z_{j,t} + r_t^\top Q r_t, \quad \sigma_t = \sqrt{\sigma_{\min}^2 + \text{softplus}(\eta_t)^2}, \quad (3)$$

$$w_{j,z} = \frac{\text{geomspace}(z_{\text{lo}}, z_{\text{hi}}, N_z)_j}{\sqrt{N_z}}, \quad Q = \frac{1}{N_r} (m_1 I + m_2 q q^\top), \quad dX_t = (\mu - \frac{1}{2}\sigma_t^2) + \sigma_t \varepsilon_t. \quad (4)$$

The same innovation drives return and reservoir. Fixing $\rho_r = -1$ and $a_r > 0$ enforces a negative instantaneous leverage direction $\kappa_0 = \rho_r a_r q^\top \sqrt{2\lambda} < 0$. This is a structural ESN mechanism rather than a fitted return/variance correlation.

The current reported architecture has $N_r = 96$, $N_z = 12$ and seven shared nonlinear-reservoir hyperparameters: z strength 0.1929, even strength 2.2291, linear strength 0.1930, gate damping 0.9651, local z coupling 0.0476, self-coupling scale 0.0397, and negative-sign probability 0.2219. Per DGP, a 12-dimensional vector—memory/rate ranges, bias and scale, rough amplitude, two z read-out endpoints, and (m_1, m_2) —is fitted by 11-start Nelder–Mead.

3 Calibration and diagnostic methodology

Each path is reduced to an 11-component stylised-fact vector: estimated Hurst exponent; mean, tail, and maximum volatility; volatility persistence; Taylor gap and fraction; Zumbach statistic; leverage; excess kurtosis; and maximum return autocorrelation. DGP cross-path means and dispersions define the target. Candidate models minimise the smooth score

$$S = \sum_{k=1}^{11} w_k [1 - f(x_k; c_k, s_k)] + 5 \text{ Stress}, \quad f(x; c, s) = \exp \left[-\frac{1}{2} \left(\frac{x - c}{s} \right)^2 \right]. \quad (5)$$

Thus $S \geq 0$ and $S = 0$ is a perfect non-stressed match. The Gaussian proximity replaced an earlier tent kernel because its gradient remains non-zero when a candidate is far from target. The protocol being studied deliberately raises the associated weight/tightens tolerance: Taylor components in the Taylor experiment, leverage in the Heston experiment, and the two Zumbach channels in the PDV experiment.

The DGPs are meaningful controlled tests. For example, roughness uses

$$\sigma_t = \sigma_0 \exp(\lambda B_t^H - \frac{1}{2} \lambda^2 t^{2H}), \quad \text{Var}[\log V_{t+\tau} - \log V_t] \propto \tau^{2H}, \quad (6)$$

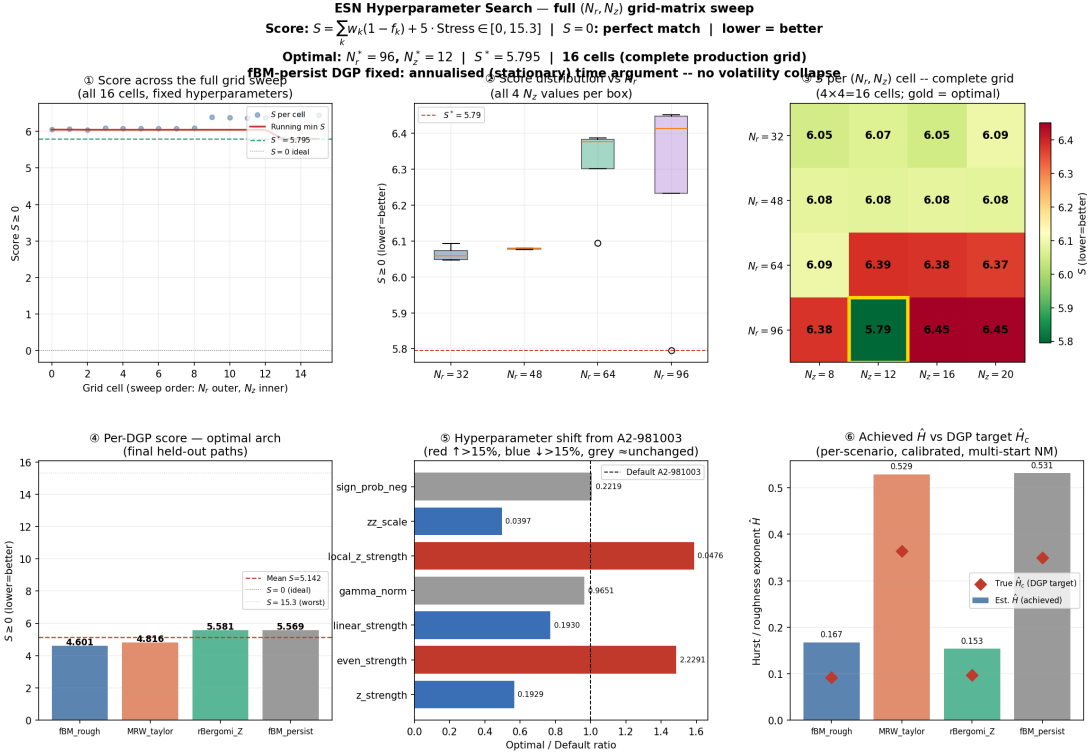
while Heston uses $dV_t = \kappa(\theta - V_t)dt + \xi\sqrt{V_t}dW_t^V$ with $\text{Corr}(dW_t^S, dW_t^V) = \rho$. The Zumbach test measures

$$Z(L) = \text{Corr}(\text{past}_L R^2, \text{future}_L V) - \text{Corr}(\text{past}_L V, \text{future}_L R^2), \quad (7)$$

so $Z(L) > 0$ signals time-reversal asymmetry. A key implementation safeguard in the fBM experiments is annualised time, $t/252$, in the covariance and Itô correction; raw day indices previously produced invalid values away from $H \approx 0$.

4 Main numerical findings

Experiment	Reported evidence	Interpretation
Architecture search	Over $(N_r, N_z) \in \{32, 48, 64, 96\} \times \{8, 12, 16, 20\}$, the reported argmin is 96, 12: $S = 5.79 \pm 0.56$ over 3 seeds.	It is only a current point estimate: the grid spread (0.66) is below typical within-cell seed noise (about 1.46).
Roughness	The ESN is calibrated at fBM targets $H = 0.05, 0.10$; scores are 3.955 and 4.216. Diagnostics include dyadic variograms, covariance scaling, and $H(q) = \zeta(q)/q$.	The experiment checks more than a scalar $\hat{H}$, but the report does not establish a production-budget, multi-seed roughness error.
Taylor	MRW targets $(G_T, F_T) = (0.086, 0.94)$ and $(0.142, 0.99)$ for $\lambda^2 = 0.1, 0.3$. ESN obtains $(0.059, 0.90)$ and $(0.043, 0.93)$.	Correct direction in both cases; close at lower intermittency, material undershoot of the larger gap. PDV-GL stays near neutral $(0, 0.51)$.
Leverage	For Heston $\rho = -1.0, -0.8$, DGP leverage is $-0.140, -0.126$; ESN reaches $-0.104, -0.092$ with scores 4.75, 4.22.	ESN captures roughly 74% and 73% of magnitude. QRH has the wrong sign; PDV-GL is about -0.006 in both cells.
Zumbach	PDV-GL is the DGP and the report decomposes both correlation terms in (7), comparing ESN and QRH lag curves.	The stated claim is mimicry of the statistic, not proof of the same causal mechanism; it is not a claim about fitting VIX-call smiles.



Saved grid-search overview: score and architecture diagnostics are generated by the repository and retained alongside the protocols.

5 Status, limits, and next validation steps

The code is a well-developed experiment harness rather than a settled empirical calibration result. It has explicit DGP sanity checks and fast modes, but no CI, package structure, pinned dependencies, or test suite. Each protocol copies common machinery, so fixes must be compared before porting. Results are simulation- and calibration-budget dependent.

Most importantly, the 96×12 architecture is not statistically separated from many tested cells at three seeds; single-run Hurst mismatches in the grid report range from 0.056 to 0.182; and the quadratic coefficients (m_1, m_2) are weakly/inconsistently identified across seeds and scenarios. The sensible next step is a longer, multi-seed, production-budget re-run at a small set of competing reservoir sizes, reporting both S and its 11-term decomposition. That

would test architecture selection, roughness matching, and whether apparent quadratic read-out effects survive calibration noise.

Sources within this repository. Root `CLAUDE.md`; `matteo/web_claude/CLAUDE.md`; five protocol scripts and their checked-in LaTeX/PDF/PNG outputs. The latter project-export material supplies research context and document orientation; this report's numerical statements are taken from the protocol reports and JSON outputs.